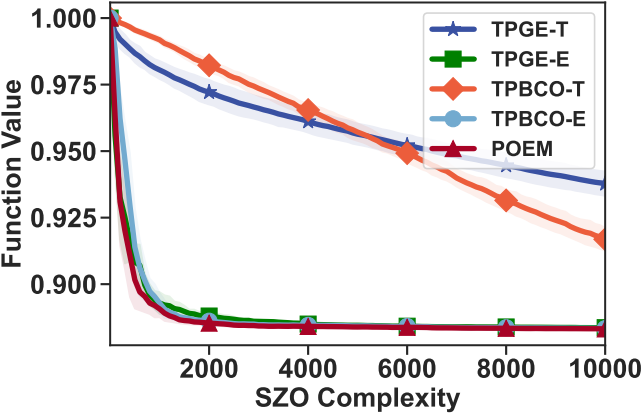
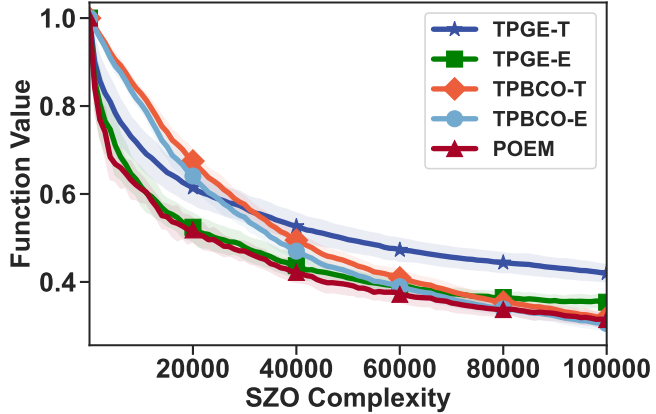


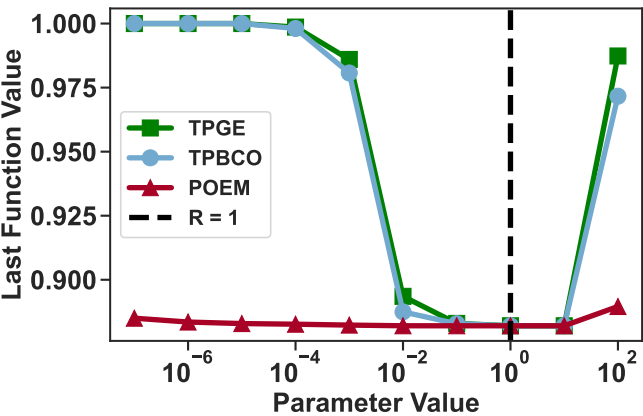


(a) qsar ( $d = 1,024, n = 1,687$ )

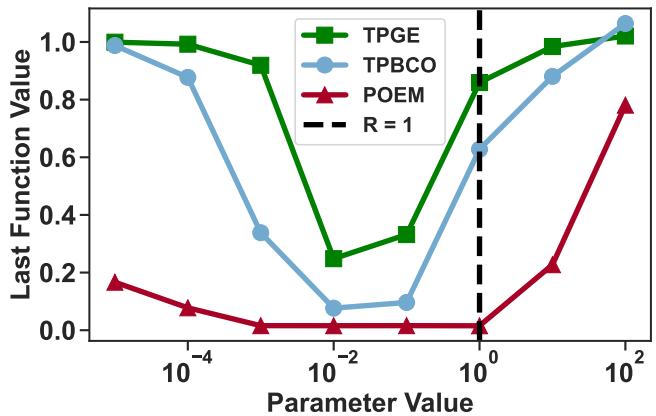


(b) gisette ( $d = 6,000, n = 5,000$ )

Figure 1. The comparison on the SZO complexity against the function value during the iterations.

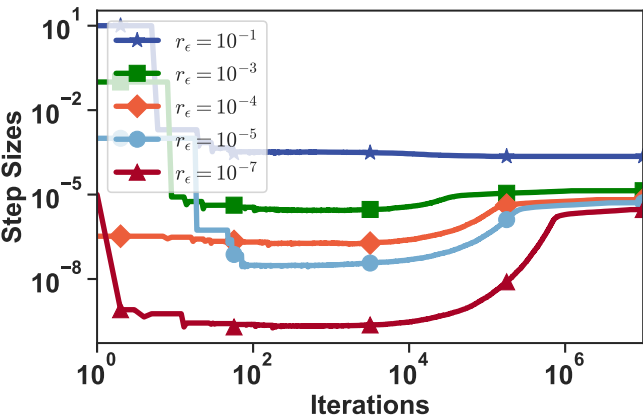


(a) qsar ( $d = 1,024, n = 1,687$ )

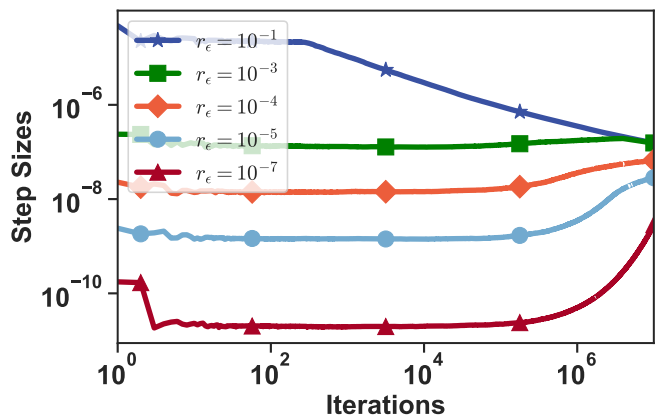


(b) gisette ( $d = 6,000, n = 5,000$ )

Figure 2. The comparison on the parameter settings ( $r_\epsilon$  for POEM and  $1/L$  for other methods) against  $f(\mathbf{x}_T)$ .



(a) qsar ( $d = 1,024, n = 1,687$ )



(b) gisette ( $d = 6,000, n = 5,000$ )

Figure 3. The change of the step size with difference  $r_\epsilon$  for POEM.